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Kirigami auxetic structure for high efficiency power harvesting in self-powered and wireless structural health monitoring systems

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Abstract

Kirigami and auxetic topologies are combined to design an innovative metamaterial-based substrate (MetaSub) for piezoelectric energy harvesters. The proposed MetaSub piezoelectric energy harvester (MPEH) contains both advantageous metamaterial properties of negative Poisson's ratio capability and enhanced planar stretchability. A computational parametric analysis is conducted to develop the optimum design for the MPEH to trap the maximum elastic energy. A finite element analysis (FEA) is employed to analytically and numerically validate the simulation model of the MPEH. Accordingly, two experimental results of conventional and auxetic strain energy harvesters are used to evaluate the power enhancement of the MPEH. The FEA results demonstrate the average power gained by the MPEH at a low level of frequency and strain excitation (10 Hz and 150 $\mu\epsilon$ peak-to-peak) is 165 μW which easily satisfies the minimum electric power amount required as a sensor node for self-powered wireless sensor networks. The harvested power output of the MPEH is 19.2 times more than power output produced by an equivalent conventional harvester with a plain substrate (8.6 μW). The performance of the MPEH is investigated at different combinations of both low and high excitation frequencies. The creative design of the MetaSub can significantly improve the productivity of strain-induced devices whose efficiency is dependent on their deformation performance such as vibration energy harvesters, wearable sensors, flexible actuators, and micro electromechanical applications.

Keywords: self-powered structural health monitoring, piezoelectric energy harvesting, auxetic, kirigami, metamaterial-based substrate, MetaSub

(Some figures may appear in color only in the online journal)

1. Introduction

Self-powered wireless sensor networks have been developed to solve the finite life-span of batteries, the difficulties of batteries replacements, the low reliability of data wires, and the high maintenance cost of power cables in structural health

monitoring (SHM) systems [1, 2]. Recently, there has been growing interest in energy harvesting from ambient mechanical vibrations due to its potential to provide an effective and sustainable supply for portable and wireless sensor networks (WSNs) of autonomous monitoring systems [3, 4]. Recently, there have been some advances in designing and fabricating

various types of vibration-based energy harvesters such as electromagnetic [5], triboelectric [6], piezoelectric energy harvesters (PEHs) [7]. Among all types of the harvesters, the strain-induced PEH has attracted a great attention over the last decade because it can perfectly harness deformation energy of structures exposed to cyclic strains, and be usable even without tuning techniques or inertia effects of any external mass [8]. As an instance, the energy provided by in-flight vibration of the aircraft's wing can be harvested by PEHs to power wireless sensor nodes for a continuous damage detection services [9]. PEHs can be also attached to traffic- or wind-induced vibration components of infrastructures such as pier, girder, stay cable, and deck to feed wireless sensors in any smart bridge monitoring systems [10]. From a biomedical standpoint, continuous healthcare monitoring demands for early detection of health disorders leads to considerable attention on feeding different kinds of compatible [11], small [12], wearable [13], or even portable [14] electronic medical sensors by PEHs [15].

Insufficient power output is the most severe limitation that makes the conventional types of PEH unusable or infeasible for the most of strain-induced applications, especially for devices subjected to a low amplitude excitation. To overcome this limitation, the MetaSub piezoelectric energy harvester (MPEH) is introduced as a PEH with a novel design of metamaterial-based substrate (MetaSub) in this paper. The MetaSub combines two key topologies of auxetic and kirigami to improve mechanical behaviors of the MPEH by the interplay between materials and specially designed geometries. A parametric study is performed to determine the optimal dimensions and properties for the MetaSub cell to achieve the maximum power output. A finite element analysis (FEA) is carried out to investigate the performance of the MPEH under a sinusoidal strain cycle, and accordingly, the feasibility of the MPEH concept is proved via finite element simulation at low and high frequencies. The MPEH concept is also computationally evaluated and compared with two experimental results of conventional and auxetic energy harvesters. The MPEH feeds sensors of WSN in any SHM systems with a highly efficient and significantly enhanced electric power output.

2. MetaSub piezoelectric energy harvester (MPEH)

Auxetic is a metamaterial concept which applies to any structure that has a negative Poisson's ratio. The Poisson's ratio defines the ratio between two characteristics of lateral and axial strain of a structure (equation (1)) [16]

$$\nu = -\frac{\varepsilon_{\text{lateral}}}{\varepsilon_{\text{Axial}}} \quad (1)$$

Structures, generally, have a positive Poisson's ratio, and contract in the orthogonal direction of applied force. However, this reaction is completely different for auxetic structures that exhibit counterintuitive deformation behavior during deformations. Indeed, when a tensile force is applied to an auxetic structure, it expands transversely instead of usual materials. The negative Poisson's ratio of the auxetic design

enables structures to be expanded in both longitudinal and transverse directions easily. In recent years, in light of the success in power output enhancement by negative Poisson's ratio, the auxetic types of energy harvesters have gradually attracted attention. One of the widespread auxetic design is the re-entrant hexagonal auxetic array shown in figure 1 [17–19]. Li *et al* [20] used an auxetic substrate to design a bimorph energy harvester with 2.76 times the power of an equivalent plain form. Power enhancement of a monomorph piezoelectric cantilever beam using a re-entrant hexagonal auxetic structure has been computationally investigated by Farhangdoust [10]. Umino *et al* [21] have generated 1.6 times more power than a plain PEH with a MEMS cantilever using an auxetic substrate. A piezo-ceramic with hexagonal honeycomb type of auxetic lattice structure has fabricated by Fey *et al* [22] to increase the strain from an applied voltage which can be used in reverse as an energy harvester.

Apart from the auxetic structure, kirigami is another metamaterial topology which comes from two words of 'kiru' (to cut) and 'kami' (paper) as a Japanese style of cutting paper [23]. The kirigami has been developed based on incised periodic and symmetric cuts which applied to shells and plates to enrich the breaking strain and deformation capacity [24, 25]. Although most of the substantial features of kirigami have recently been introduced in various engineering applications [26, 27], the use of the kirigami concept in PEHs is still at an early research stage. As one of the most widely discussed kirigami topology, the re-entrant uniaxial kirigami design is illustrated in figure 1 [26, 28]. Sun *et al* [29] experimentally proved a 2.6 times improvement in voltage output, obtained from a soft and highly deformable PEH with a kirigami-based substrate. The kirigami-based PEH has been proposed for biomedical applications, with a soft and highly deformable substrate like polydimethylsiloxane (PDMS) or polyethylene terephthalate (PET), and this model is subject to cracking as well as transferring in-plane deformation to out-of-plane deformations in other engineering applications which use a more rigid substrate like steel [30, 31].

MetaSub is an innovative design for PEH substrate which combines two key metamaterial-based concepts of auxetic and kirigami (figure 1). The MetaSub adds the negative Poisson's ratio capability to the kirigami substrate. In other words, the MetaSub applies kirigami cuts to an auxetic framework to provide MPEH with a remarkable multi-directional expansion and adequate deformation, compared to the conventional PEH.

The MetaSub unit cell configuration with details is shown in figure 2. This design is based on a network of connected four triangles with two opposing kirigami cuts pushed inwards. When this structure is placed under tension, the two kirigami cuts open up and the triangles push outward. More specifically, the bowtie shape tends toward rectangular as the first through fourth triangle elements flex outward in opposite transverse directions.

PEHs usually use one of the operating modes of 31, 33 or 15 to generate electric power. In this study, the d_{31} operation mode of polarization was considered in which the piezoelement (PZT) is poled along the 3-axis and the excitation force is applied along the longitudinal direction (1-axis). The

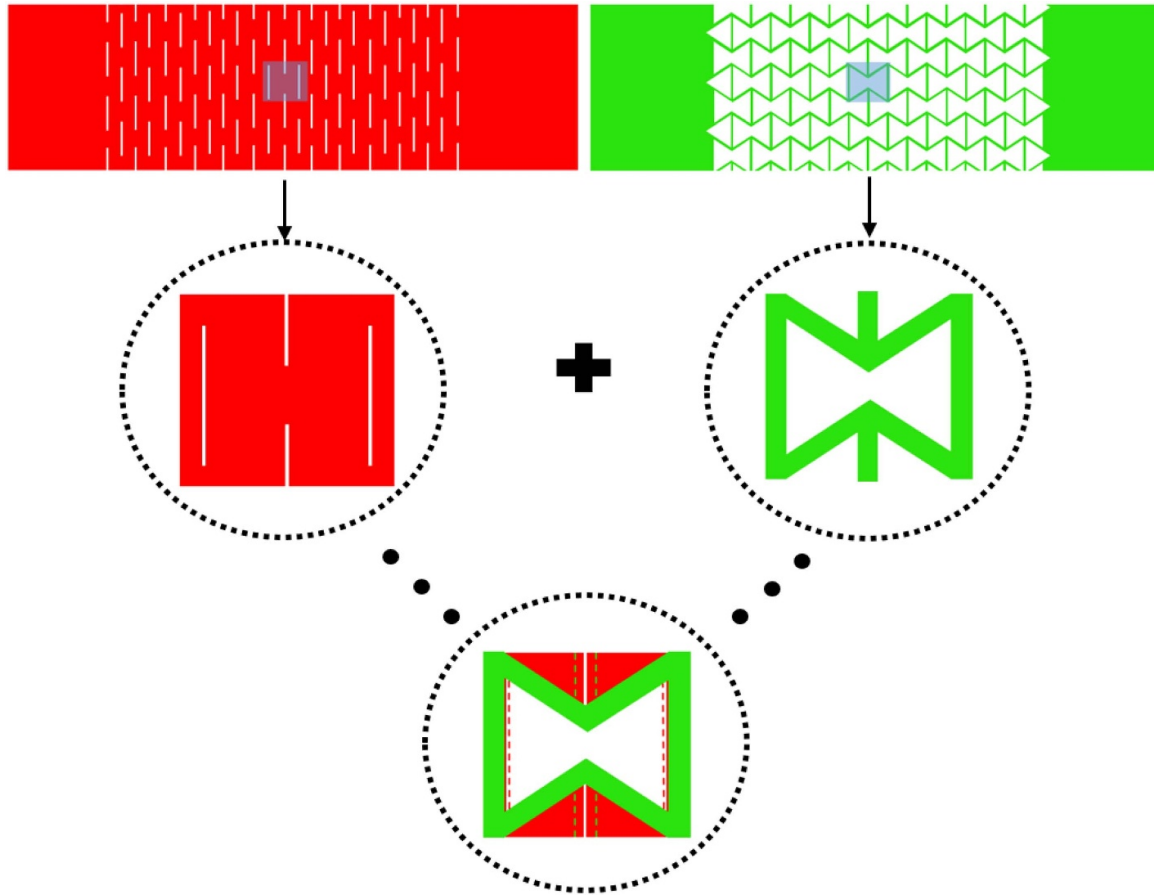


Figure 1. The MetaSub cell design (bottom) combining re-entrant hexagonal auxetic (green) and uniaxial kirigami (red) topologies.

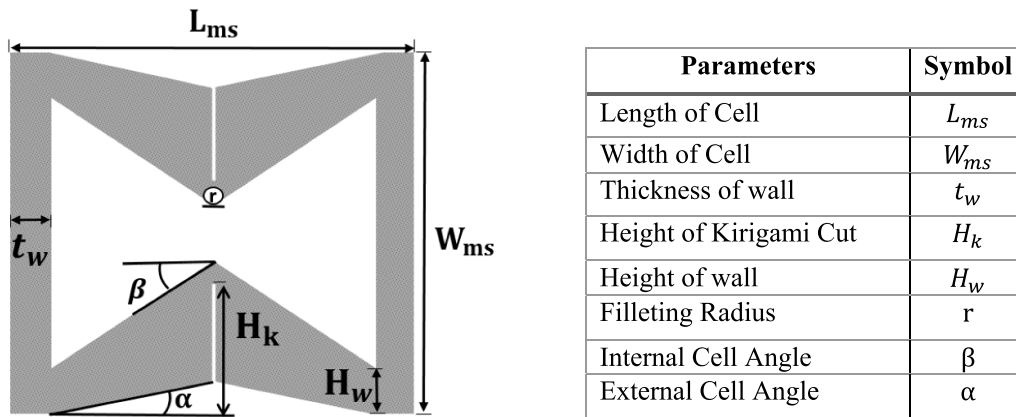


Figure 2. Schematic drawing showing the design of the MetaSub unit cell with details.

MetaSub provides the MPEH a high tensile deformation capacity compared to the conventional PEH with a plain substrate. From the structural analysis's standpoint, unreformed and deformed states for two same size of plain and MetaSub cells are shown in figure 3.

In general, a substrate transfers the strain energy of the applied excitation over the PZT and thus polarizes the PZT to generate electric power output. As shown in figure 3, the plain substrate transfers an axial stress of σ_{11} and a lateral stress of

σ_{22} over the PZT when this structure is placed under tension along the 1-axis. However, when a tensile force is applied to the MetaSub, the PZT experiences a different stress distribution, with stress concentrations consisting of σ_A and σ_K related to the auxetic framework and kirigami cuts, respectively. With the improved features of kirigami and auxetic applied to the MetaSub cell design, a significantly enhanced power output is expected from any PEH using the MetaSub as its substrate for two reasons:

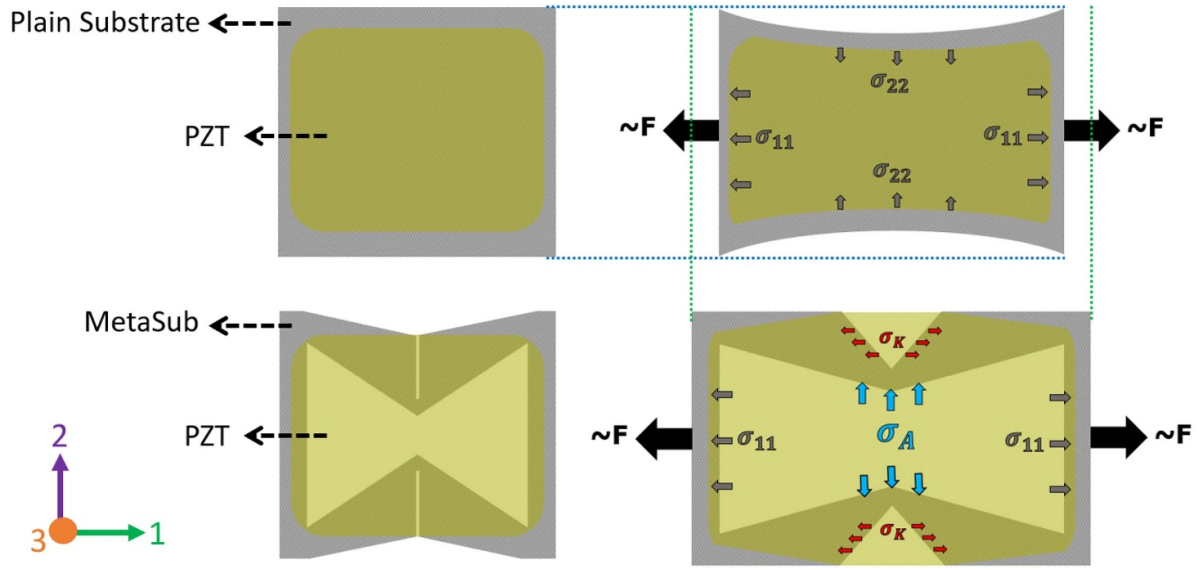


Figure 3. Undeformed (left) and deformed (right) states of the MetaSub cell (bottom) and plain cell (top).

- The MetaSub stretches the PZT in two directions at once due to its auxetic negative Poisson's ratio which produces stress concentrations (σ_A) in the PZT (figure 3), and thus increases the electric power output.
- The MetaSub behaves more flexibly with enhanced strain capacity owing to its kirigami cuts, which produces stress concentrations (σ_K) in the PZT (figure 3), and accordingly boosts the polarization to generate more power.

The expected reactions of the two MetaSub and plain unit cells and their corresponding effects as the substrate on power output of PEH are validated in following sections.

3. Validation

Piezoelectric (PZT) material consists of a network of asymmetric dipole crystals which polarized when a mechanical stress is applied to the structure. This network enables the PZT to generate electrical potential in direct proportion to the applied forces. The constitutive equation for the electromechanical characteristic of a linear piezoelectric sensor is defined as [32, 33]:

$$\begin{bmatrix} \text{Converse} \\ \text{Direct} \end{bmatrix} = \begin{bmatrix} S \\ D \end{bmatrix} = \begin{bmatrix} k^E & d^t \\ d & \varepsilon^T \end{bmatrix} \begin{Bmatrix} \sigma \\ E \end{Bmatrix} \quad (2)$$

where D , σ , k^E , and ε^T denote, respectively, the dielectric displacement vector (the electrical charge density), applied stress vector, compliance under a constant electrical field, and the dielectric permittivity matrix at constant stress. S , d , d^t , and E also represent the strain, direct piezoelectric effect matrix, converse piezoelectric effect matrix, and electric field, respectively. The following alternative form of piezoelectric constitutive equations is used for direct piezoelectric effect of PEHs [34]:

$$S_{ij} = k_{ijkl}^E T_{kl} + d_{kij} E_k \quad (3)$$

$$D_i = d_{ikl} \sigma_{kl} + \varepsilon_{ik}^T E_k \quad (4)$$

It is assumed that the excitation force is applied along the longitudinal direction (X -axis). Accordingly, the PZT will be polarized along the thickness direction (Z -axis) (figure 4), and then the piezoelectric constitutive equations will be expressed as [35]:

$$S_1 = k_{11}^E T_1 + d_{31} E_3 \quad (5)$$

$$D_3 = d_{31} \sigma_1 + \varepsilon_{33}^T E_3 \quad (6)$$

Assuming an isotropic behavior and a plain stress condition ($\sigma_{33} = 0$) for the PZT, the electric displacement in axis 3 can be reduced to the following scalar equation (7) [36]:

$$D_3 = d_{31} (\sigma_{11} + \sigma_{22}) \quad (7)$$

in which the piezoelectric constant d_{31} denotes the charge per electrode area per unit stress applied along axis 1. A circuit model representing a strain piezoelectric energy harvester (SPEH) connected to a load resistor is depicted in figure 5. The electric current output of the piezoelectric layer can be obtained from the integral of Gauss's law as [37]:

$$\frac{d}{dt} \left(\int D \cdot n dA \right) = \frac{v(t)}{R_L} \quad (8)$$

Where the V_{oc} , C_p , and R_L , respectively, are the open-circuit voltage, parasitic capacitance between the electrodes on two sides of the PZT, and load resistance of the SPEH with a thickness of t_p . As the PZT ceramic is a bulk structure ($d_{31} = d_{32}$), when an uniaxial stretching load is applied to the SPEH, the root mean square (RMS) of the voltage for electrical circuit of

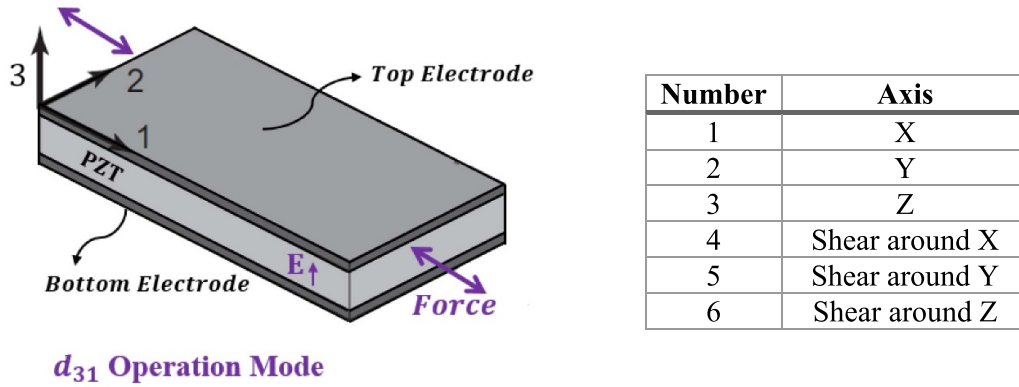


Figure 4. The schematic of a piezoelectric element, with directions.

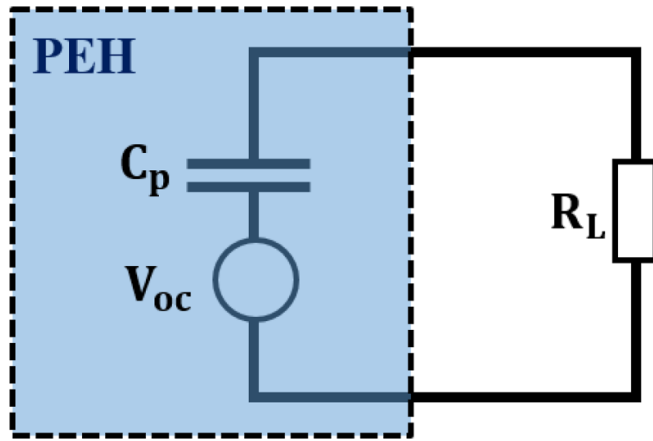


Figure 5. The equivalent circuit model of a SPEH connected to a load resistor.

a thin PZT layer under dynamic bending will be calculated by [20, 38]:

$$V_{OC} = \frac{t_p D_3}{\varepsilon_0 \varepsilon_{33}} = \frac{t_p d_{31}}{\varepsilon_0 \varepsilon_{33}} (\sigma_{11} + \sigma_{22}) \quad (9)$$

$$V_{RMS} = \frac{V_{OC}}{\sqrt{2}} \quad (10)$$

For the SPEH, power output is proportional to the square of the RMS of the voltage output. The maximum power output (equation (12)) of the SPEH takes place at the optimal load resistance (equation (11)) which matches the internal impedance of the PZT [39]

$$R_L = \frac{1}{2\pi f C_p} = \frac{t_p}{2\pi f \varepsilon_0 \varepsilon_{33} A_p} \quad (11)$$

$$P_{opt} = \frac{V_{RMS}^2}{R_L} = \frac{f A_p t_p d_{31}^2}{\varepsilon_0 \varepsilon_{33}} (\overline{\sigma_{11}} + \overline{\sigma_{22}})^2 \quad (12)$$

in which, ε_0 and A_p denote permittivity of free space and electrode area of the PZT, respectively. The factor of $d_{31}^2/\varepsilon_{33}$ is

known as the energy harvesting figure of merit (FOM) which effectively describes the change in stored power output within the PZT when a stretching load is applied to the SPEH [40]. The FOM is usually considered as the key material properties for effective PZT selection [41].

Furthermore, as it is expressed by equation (12), it can be also shown that the extracted power from the PZT of the SPEH is proportionally related to the square of the trace of mean axial and lateral stress tensors (equation (13))

$$P_{opt} \propto (\overline{\sigma_{11}} + \overline{\sigma_{22}})^2 \quad (13)$$

To validate the concluded physical interpretation (equation (13)), an FEA was employed to simulate and evaluate the 3D model of the MPEH by COMSOL Multiphysics. Accordingly, the performance of the MPEH was compared with two experimental results of a conventional PEH and an auxetic piezoelectric energy harvester (APEH). In simulation, the MPEH model comprises a PZT layer bonded to the MetaSub cell by a thin elastic layer (TEL) of glue (figure 6).

The substrate has two clamped regions at its ends, each with a length of L_{clp} which provide different boundary conditions to the model. One clamped region is fixed, and another one is subjected to an axial harmonic displacement to produce an applied excitation. The dimensions and preferable values of the model are listed in table 1. The material properties of different parts of the MPEH including PZT, substrate, and glue are listed in table 2.

As the first case study, to demonstrate the significant power enhancement of the MPEH, an equivalent conventional PEH using a plain substrate was also modeled. Then, to prove the validity of the MPEH output, the both simulated models were evaluated to experimental results of the conventional PEH [42]. For this performance evaluation, material properties (table 1) and design parameters (table 2) were selected similar to the experimental test [42]. To this end, excitation frequency, glue TEL, and amplitude sinusoidal strains were taken, respectively, 10 (Hz), 173 (GN m⁻³), and 250 ($\mu\varepsilon$). Similarly, to investigate the simulated models in an identical condition to experimental tests, the PIC151 was selected to be considered as the PZT, however according to equation (12) it

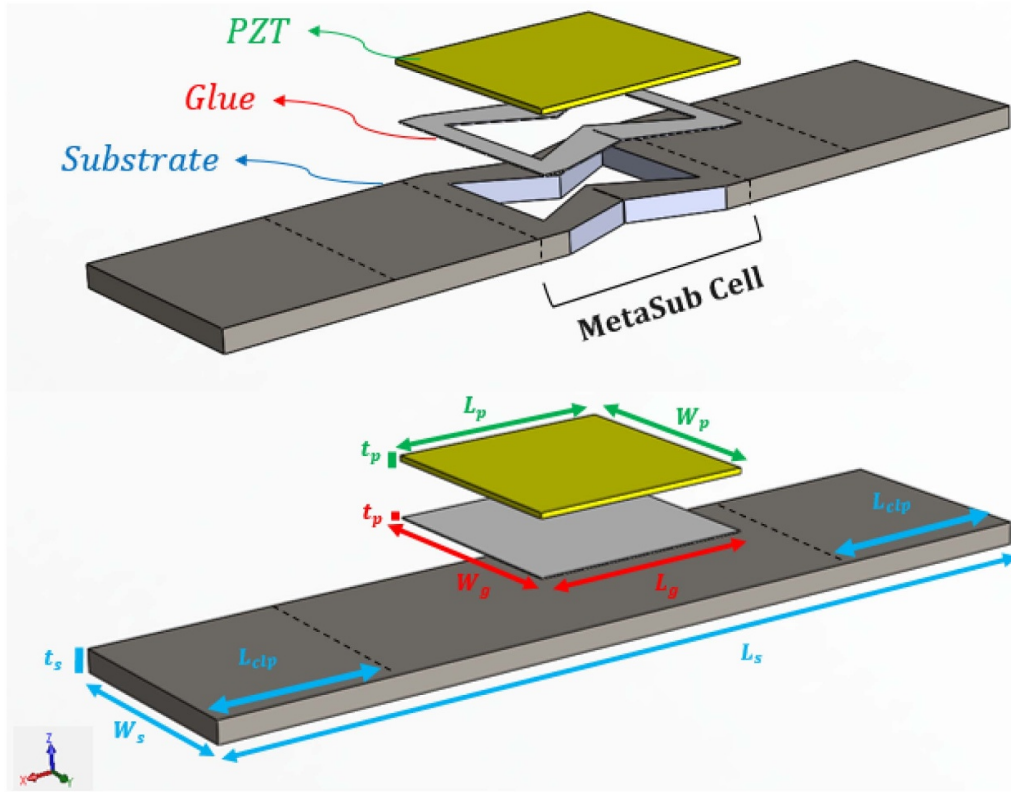


Figure 6. The MPEH (top) and the conventional PEH (bottom).

Table 1. Parameter values used in the models.

Part	Parameter	Value	Unit
Substrate	Length, L_s	80	mm
	Width, W_s	20	mm
	Thickness, t_s	1.15	mm
	Clamp, L_{clp}	20	mm
MetaSub cell	Length, L_{ms}	20	mm
	Width, W	20	mm
	Thickness of wall, t_w	2	mm
	Height of kirigami cut, H_k	6	mm
	Height of wall, H_w	2	mm
	Filleting radius, r	0.4	mm
	External angle, β	40	degrees
	Internal angle, α	5	degrees
	Length, L_g	20	mm
	Width, W_g	20	mm
Glue	Thickness, t_g	0.08	mm
	Length, L_p	20	mm
	Width, W_p	20	mm
Piezo-element	Thickness, t_p	0.18	mm

might not be the best PZT candidate because of its high piezo permittivity (e.g. $\epsilon_{33}^T = 2109$) which may result in low FOM.

As it is shown in figure 7, the average power output curve of the simulated model is in a good agreement with the experimental results. At the electrical load resistance of 463 (k Ω), the volume of 13.4 μ W was experimentally recorded as the peak power for the conventional PEH [42] which satisfactorily compares to the power output of our simulated PEH (14.8 μ W) at the corresponding electrical load resistance.

What is impressive for this case study is the maximum power output of the MPEH, 193.8 μ W, which demonstrates a significant enhanced power output compared to the conventional PEH. The power output is not for optimized MPEH yet, and it can be further improved after a parametric design study as in section 4.

Recently, APEH has been introduced as a potential solution for increasing the power output of the conventional PEH. Experimental results showed that an APEH using a re-entrant hexagonal unit cell is able to produce electric power 14.4 times more than the conventional PEH [42]. As the second case study of the model verification, the proposed MPEH was compared to the experimental results of the APEH [42]. To this end, the excitation frequency, glue TEL, and peak-to-peak sinusoidal strains, respectively, were considered to be 10 (Hz), 220 (GN m $^{-3}$), and 250 ($\mu\epsilon$) similar to the experimental test.

Table 2. Material properties for PEHs.

Material	Property		Value	Unit
Piezo-element: PZT-PIC 151	Density	ρ_p	7500	kg m^{-3}
		s_{11}^E	16.83	$\mu\text{m}^2 \text{N}^{-1}$
		s_{33}^E	19.00	$\mu\text{m}^2 \text{N}^{-1}$
		s_{55}^E	50.96	$\mu\text{m}^2 \text{N}^{-1}$
		s_{12}^E	-5.656	$\mu\text{m}^2 \text{N}^{-1}$
	Compliance matrix	s_{13}^E	-7.107	$\mu\text{m}^2 \text{N}^{-1}$
		s_{44}^E	50.96	$\mu\text{m}^2 \text{N}^{-1}$
		s_{66}^E	44.97	$\mu\text{m}^2 \text{N}^{-1}$
		d_{31}	-214.0	pC N^{-1}
	Coupling matrix	d_{33}	423.0	pC N^{-1}
		d_{15}	610.0	pC N^{-1}
		ε_{11}^T	1936	—
	Relative permittivity (at constant stress)	ε_{33}^T	2109	—
Substrate: Steel-EN 10130	Density	ρ_s	7870	kg m^{-3}
	Poisson's ratio	ν_s	0.29	—
	Young's modulus	E_s	200	GPa
	Density	ρ_g	1250	kg m^{-3}
Glue:	Poisson's ratio	ν_g	0.35	—
	Young's modulus	E_g	1.00	GPa

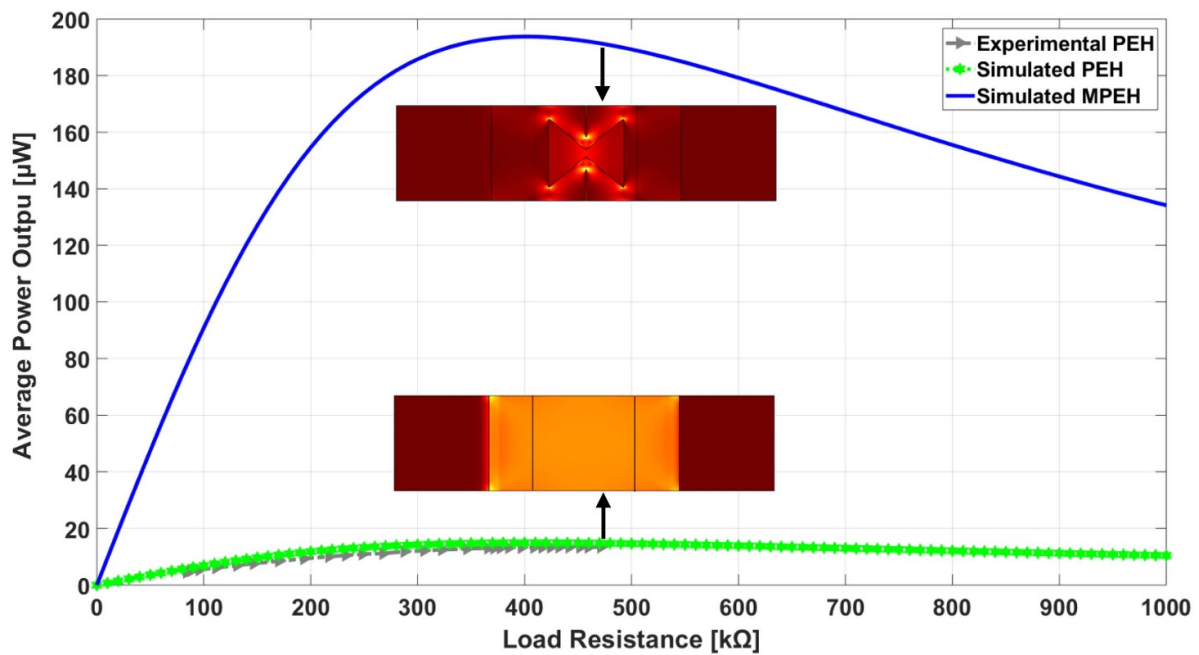
**Figure 7.** Average power output against load resistance for experimental conventional PEH [42], simulated conventional PEH, and simulated MPEH. The specimens are shown from a top view.

Figure 8 shows a great agreement of our simulated APEH with the experimental results. As a result of this FEM analysis, at the electrical load resistance of 360 (k Ω), the volume of 191.1 μW was, experimentally, recorded as the peak power for APEH [42], which satisfactorily matches the power output of the simulated APEH (196.9 μW) with an error of less than 3% at the corresponding electrical load resistance. Besides, for this case study, the maximum power output of the MPEH

is 270.5 μW , which is a greatly improved power output compared to APEH. Simulation results show that the MPEH can generate 1.41 times more power output than the APEH. The power output is not for optimized MPEH yet, and it will be more improved after a parametric design study as discussed in section 4.

Apart from those proofs of the MPEH concept, the maximum power outputs for the simulation model of the

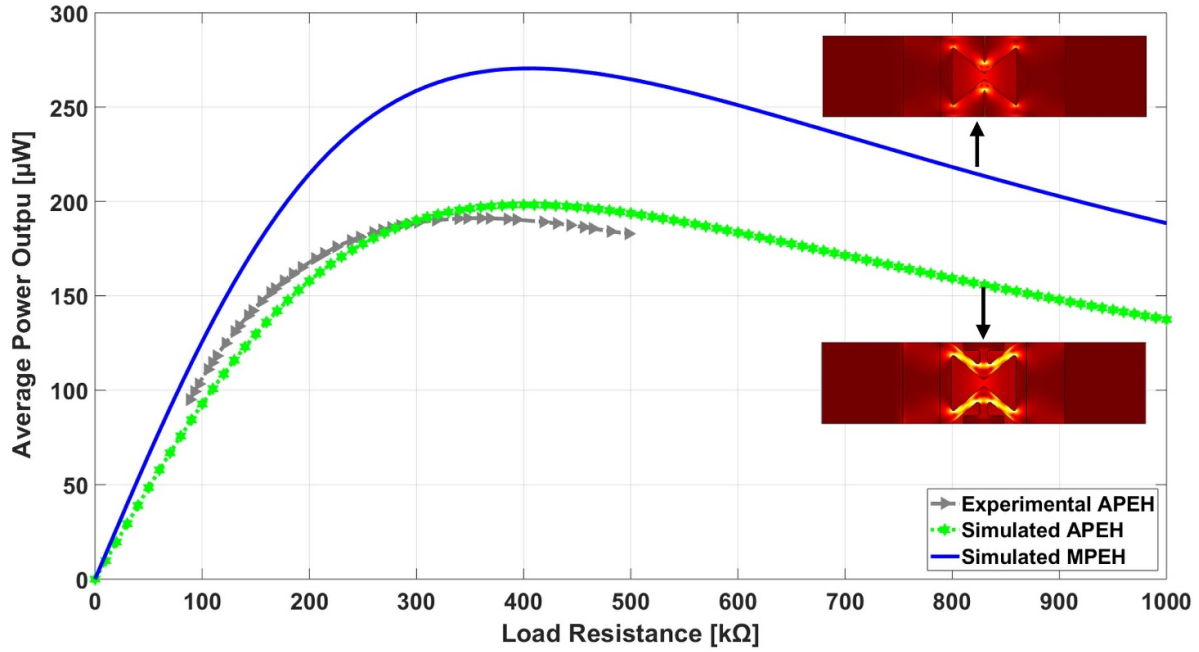


Figure 8. Average power output against electrical load resistance for experimental APEH [42], simulated APEH, and simulated MPEH. The specimens are shown from a top view.

Table 3. Comparison of the mean stress and power output values for both PEH and MPEH.

Piezoelectric energy harvester	Square of mean stress value	Average of optimum power output
Conventional PEH	7.02 MPa	8.6 μ W
MPEH	134.09 MPa	165 μ W

MPEH in both validation cases was observed at 400 (k Ω), which matched with what we expected from equation (11) for the optimal value of the load resistance. Furthermore, as it is shown in figures 7 and 8, the power output first increases with the load resistance and then gradually stabilizes.

To validate the concluded physical interpretation from equation (13), the conventional PEH and the MPEH were subjected to an axial harmonic displacement to produce a strain of 150 $\mu\epsilon$ peak-to-peak at a frequency of 10 Hz. At this low excitation condition, the stress and harvested power values for the FEM models are compared in table 3. As a result of FEM analysis, the average of optimum harvested power for the conventional PEH and MPEH, respectively, are 165 μ W and 8.6 μ W, which are in excellent agreement with equation (13), which says the greater the square mean stress value is, the greater the harvested power output will be.

4. Parameter optimization

Once the MPEH model was validated, we operated a parametric analysis of the MetaSub to identify the optimum dimensions, as judged by their ability to generate the highest power output without excessive stress. The key parameters defining

the shape of MetaSub cell are the external cell angle, α , the kirigami cut length, h_k , and the filleting radius in all the corners, r .

In this parametric study, when each of the aforementioned parameters was varied, the others were kept constant. For all the simulations, the excitation frequency and peak-to-peak sinusoidal strains, respectively, were considered to be 10 (Hz) and 250 ($\mu\epsilon$). The values used in each step are the optimum ones obtained in the previous step.

4.1. Kirigami cut length (h_k)

A range of 0.2–6.8 was selected to be studied as the preferable range for the kirigami cut length. Two kirigami cuts were added to the auxetic framework to enrich the lateral strain capability of the MetaSub cell and thus aid the lateral strain transfer which causes the kirigami stress distribution (σ_K) into the PZT. From figure 9, the maximum power output may be obtained when the length of the kirigami cut is 6.6 mm. Furthermore, utilizing a kirigami length more than 6.6 mm would be infeasible, as the space between the end of the cut and the central corner of the bowtie would be too narrow and cause crack propagation due to too high a high stress concentration.

4.2. External cell angle (α)

The MetaSub cell itself may be defined by the cell angle, α , illustrated in figure 10. It is related to how far the unit could potentially expand outwards when a stretching load is applied to the MPEH. This MetaSub parameter plays an important role in connecting the kirigami cuts to the auxetic framework, and in the expansion ability of the MetaSub cell. Accordingly, the

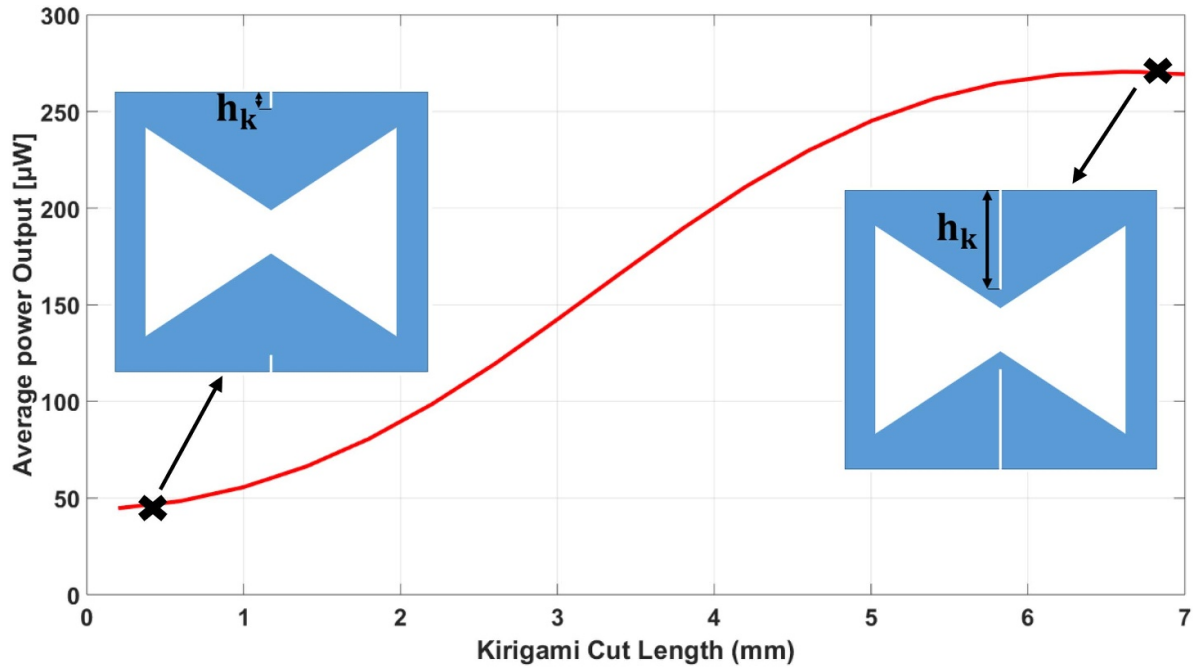


Figure 9. Power output for different kirigami cut lengths, h_k .

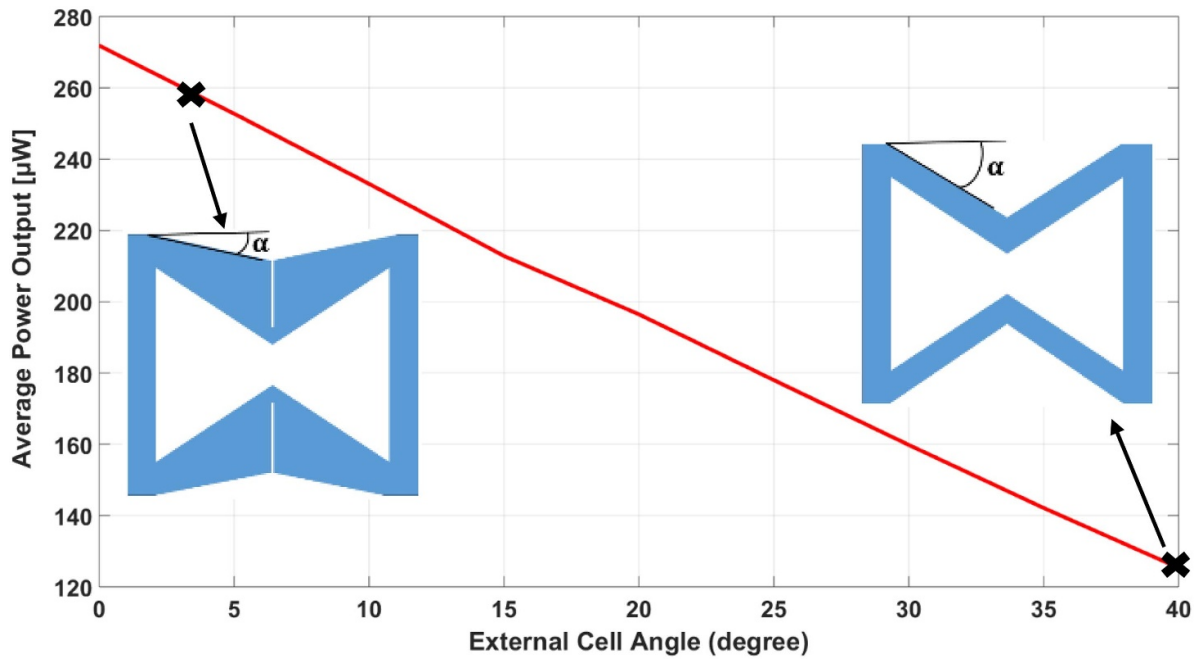


Figure 10. Power output for different external cell angle, α .

more the MetaSub region is able to expand under the stretching load, the more lateral strain (and thereby additional power) in the PZT. To amplify the auxetic behavior of the MetaSub, a range of 0–40° was selected for the external cell angle. In the undeformed state, the external cell angle can be equal or smaller than the internal cell angle. As it is shown in figure 10, the optimum for power output occurs when the external cell angle is zero when the kirigami cuts was kept constant at 6.6 mm

4.3. Filleting radius (r)

Figure 11 shows the effect of filleting the corners, ranging from the minimum radius (0.2 mm) up to the maximum radius (1.3 mm), on the power output of the MPEH. When the filleting radius is large, the extra material in the corners impedes the flexing of the MetaSub cell, and accordingly the harvested power output is decreased. As shown in figure 11,

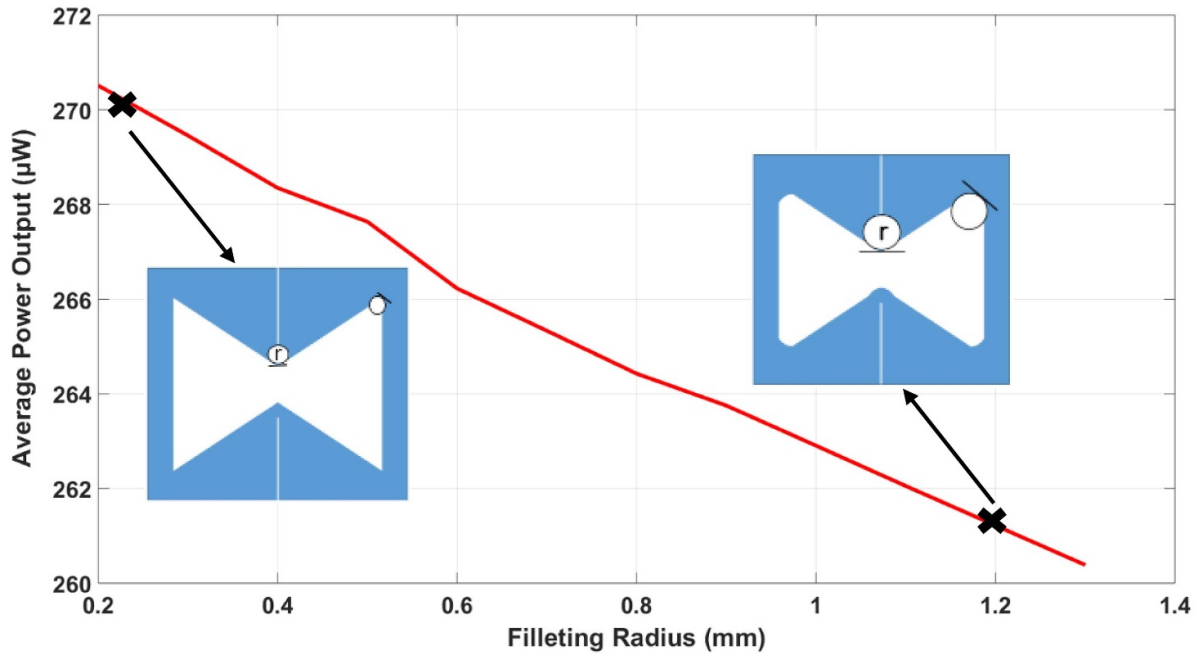


Figure 11. Power output for different filleting radius, r .

the power output almost linearly decreases when filleting radius increases. For the filleting radius, 0.2 was obtained as the optimum value for power generation.

According to the design parametric study performed in this section, table 4 shows the optimum values of the selected properties for the MPEH.

5. Results and discussion

Although different types of PEH have been introduced so far, harvesters which can generate a significant power at a low level of frequency and strain amplitude of excitation are scarce. In this section, finite element model (FEM) simulation using COMSOL Multiphysics 5.3a was carried out to demonstrate the remarkable power enhancement and high efficiency of the MPEH in power generation at low levels of excitations. For this computational study, a frequency of 10 Hz, PZT's thickness of 80 (μm), and strain amplitude of 150 $\mu\epsilon$ peak-to-peak (i.e. a peak extension of 6 μm) were selected as the excitation of the MPEH. The applied resistance load was selected as 190 k Ω based on equation (11). The same material properties and design parameters used in tables 1 and 2 were selected, including optimized parameters of table 4.

As it was mathematically proven earlier, the power output performance of any PEHs depends on the square sum of the lateral and axial stresses of its attached PZT. Here, the FEM analysis clearly shows the significant difference between the stress value of the PZT used in the MPEH and the conventional equivalent PEH harvester. The average stress value of the PZT in axial (σ_{11}) and lateral (σ_{22}) directions for the MPEH are 7.37 MPa and 4.21 MPa, respectively. However, the average stress value of the PZT in axial and lateral directions for the plain PEH are 3.12 MPa and -0.47 MPa, respectively. Table 5

Table 4. Optimized design parameters for the MetaSub cell.

Parameter	Optimized value (study range)	Unit
Kirigami cut length, h_k	6.6 (0.2–6.8)	mm
Filleting radius, r	0.2 (0.2–1)	mm
External cell angle, α	0 (0–40)	degree

Table 5. Comparison of the mean stress values for both PEH and MPEH.

Piezoelectric energy harvester	Axial mean stress value, (σ_{11})	Lateral mean stress value, (σ_{22})
MPEH	7.37 MPa	4.21 MPa
Conventional PEH	3.12 MPa	-0.47 MPa

summarizes the computational results. The MPEH with generated peak power of 165 μW presents a power multiplication of 19.2, compared to the electric power of 8.6 μW produced by the equivalent plain harvester.

The result of table 5 logically proves that the distinctive geometry of the MetaSub cell was the rational reason of the power enhancement. Then, to verify the mechanism of the power increase in MPEH, the stress distribution across the PZT in the both MPEH and plain harvesters were investigated, as shown in figure 12. Furthermore, the expected reaction from the MetaSub mechanism (figure 3) can be clearly observed in the false color image of the PZT in the figure 12.

Furthermore, another indication of the efficiency of the MetaSub concept is the stress concentration across the PZT

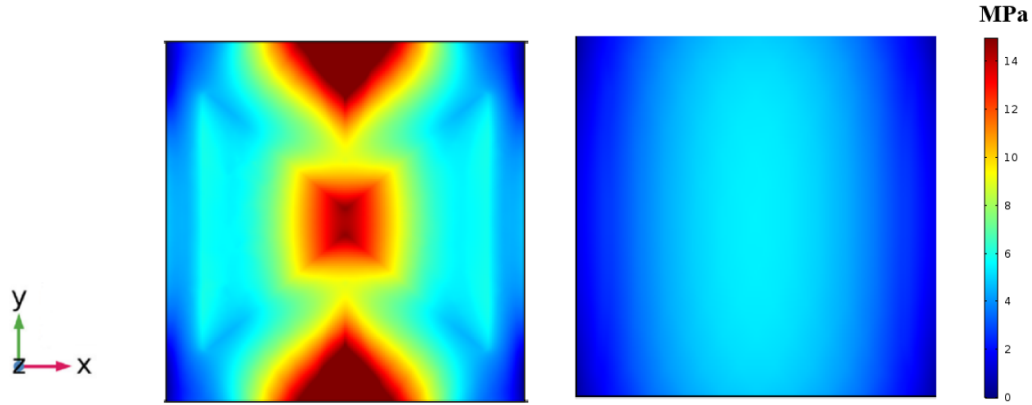


Figure 12. The PZT stress distributions of the MPEH (left) and conventional PEH (right).

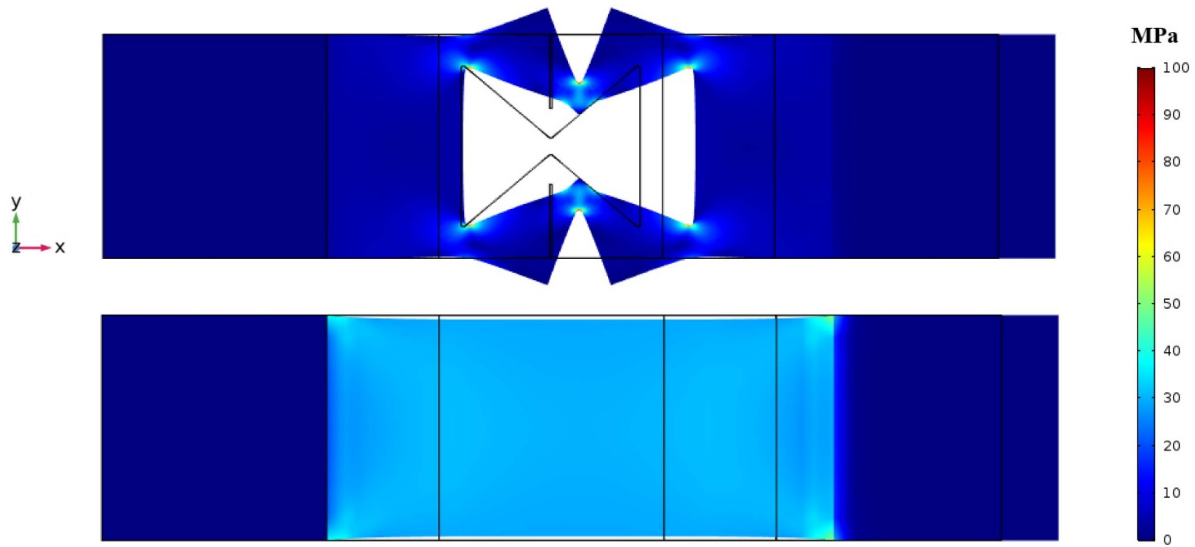


Figure 13. The displacement and stress distribution for plain substrate (bottom) and MetaSub (top). The deformation is scaled up by 850 times, and the black lines denote the original shape.

used in MPEH, is much greater than the equivalent conventional PEH. The max stress across the PZT is 28.9 MPa which demonstrates the MetaSub design can use the 82.6% of PZT capacity (with tensile strength 35 MPa) in generating power. This great performance gives the MPEH a significant efficiency compared to the conventional PEH, with a max stress of 4.9 MPa, using only 7.1% of the PZT capacity.

Figure 13 illustrates the displacement of the MetaSub and plain substrate along the X and Y axes. As the substrate of the MPEH is stretched in the X -axis, a remarkable expansion is observed for the MetaSub unit cell in the Y -axis which demonstrates its negative Poisson's ratio behavior. However, when the plain substrate is stretched longitudinally, it is transversally contracted which is typical for a conventional substrate with a positive Poisson's ratio. Figure 13 also shows that the stress distribution across the MetaSub is at a very low level in comparison with the plain substrate.

As it is shown in figure 13, the plain substrate of the conventional PEH has a uniform stress distribution in the area between the two clamped regions. In contrast, in the MetaSub cell, the stress is concentrated into corners, particularly into

those corners that flex outward. These corners become more acute due to the reduced stiffness, and thus we acknowledge that all the corners are vulnerable to excessive concentration of stress. However, the maximum stress concentration of the MetaSub cell is the stress of 198 MPa, caused by kirigami cuts found at the apex of the corners. This stress is sufficiently below the yield strength (280 MPa) of the steel substrate.

To predict a magnification factor for the MPEH power harvesting, its power output was investigated for different excitation frequencies and amplitude strains. One of the clamped regions of the substrate was modeled as fixed, and another one was subjected to an axial harmonic displacement, to produce a strain range of 100–550 $\mu\epsilon$ at a frequency range of 1–16 Hz. The effects of the strain and frequency on the performance of the MPEH are shown in figures 14 and 15. As a result, the average power output will be increased when either the applied frequency or amplitude is increased.

When the frequency was varied, the strain amplitude was kept constant at 150 $\mu\epsilon$ peak-to-peak (figure 14). Similarly, the applied frequency was kept constant at 10 Hz, when the strain amplitude was varied (figure 15).

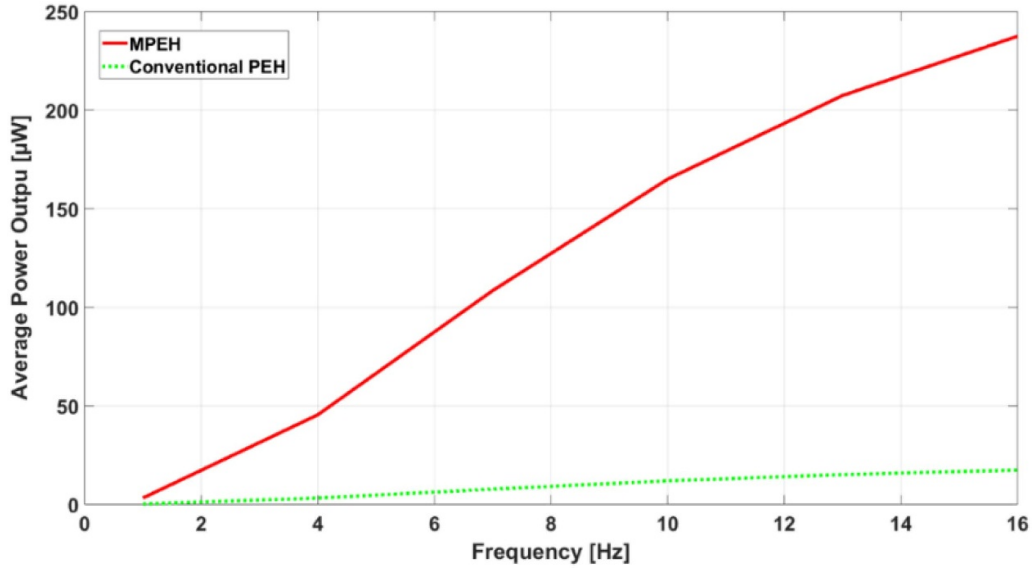


Figure 14. Average power output of both MPEH and conventional PEH at different frequency.

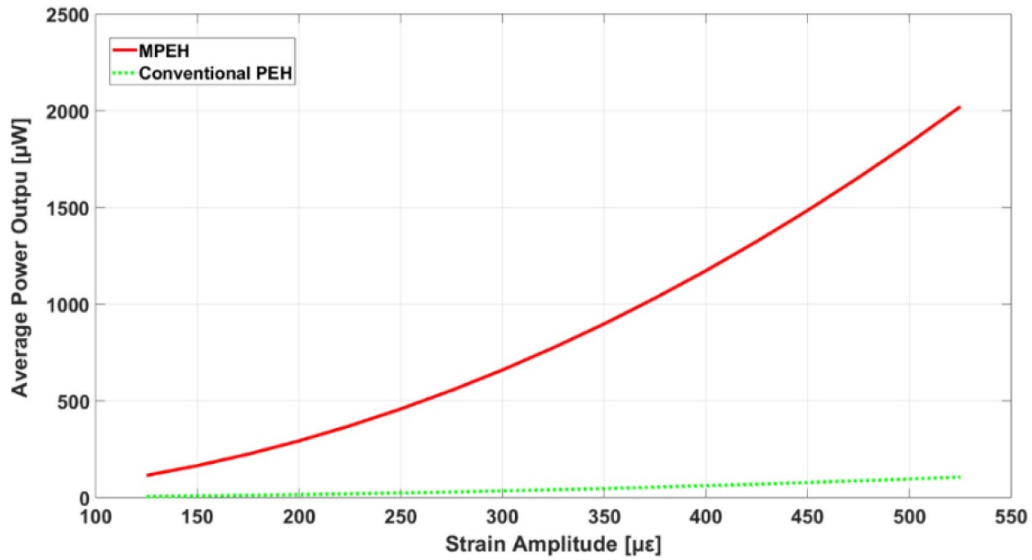


Figure 15. Average power output of both MPEH and conventional PEH at different amplitude.

As it shown in figure 16, the MPEH's response was investigated at low excitation frequencies range of 1–16 Hz. The contour illustrates that the average power outputs of the MPEH significantly increase with frequency and strain amplitude.

To analyze the performance of the MPEH at a broader frequency bandwidth, different combinations of frequency–amplitude were investigated at a frequency range of 100–1500 Hz. Figure 17 shows the MPEH's response for different combinations at high excitation frequencies.

In order to compare the harvesting performance of the MPEH against the conventional PEH, a comparison ratio η is defined as the power gain factor (equation (14))

$$\eta = \frac{\text{Power}_{\text{MPEH}}}{\text{Power}_{\text{PEH}}} \quad (14)$$

Figure 18 shows the distribution of the power gain factor for MPEH as the applied amplitude and frequency are varied. It was demonstrated that 19.3 can be reached as the magnification factor for some combinations of strain and frequency at low level excitations.

As it is shown in figure 19, the power gain factor was also investigated at a broader frequency bandwidth (100–1500 Hz). It is shown that a magnification factor of 18.8 can be reached as the magnification factor for high excitation frequency range. This obviously demonstrates the power gain factor remains constant when the strain amplitude of the excitation is increased, indicating the magnification factor is a function of the substrate geometry of the MPEH, and is not dependent on the excitation conditions.

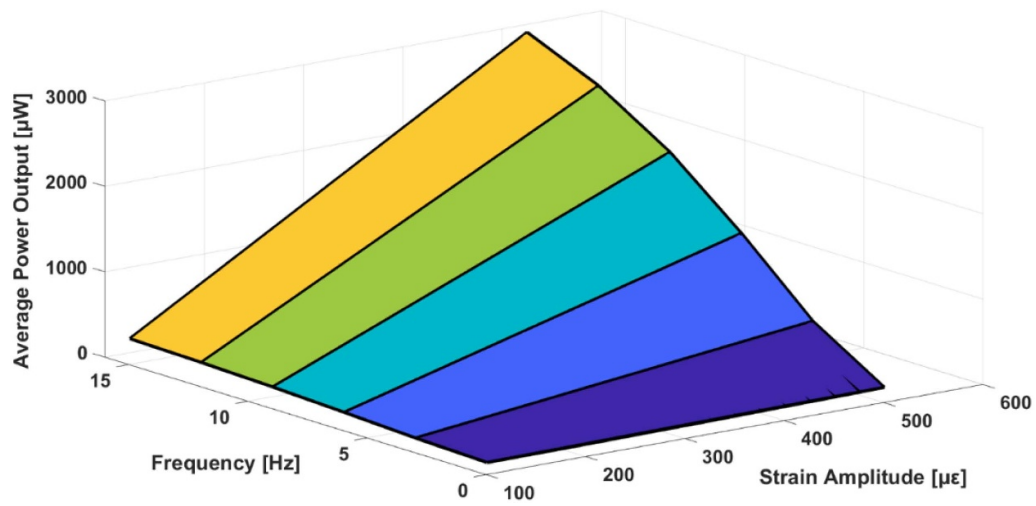


Figure 16. Average power output contour for the MPEH at low frequencies.

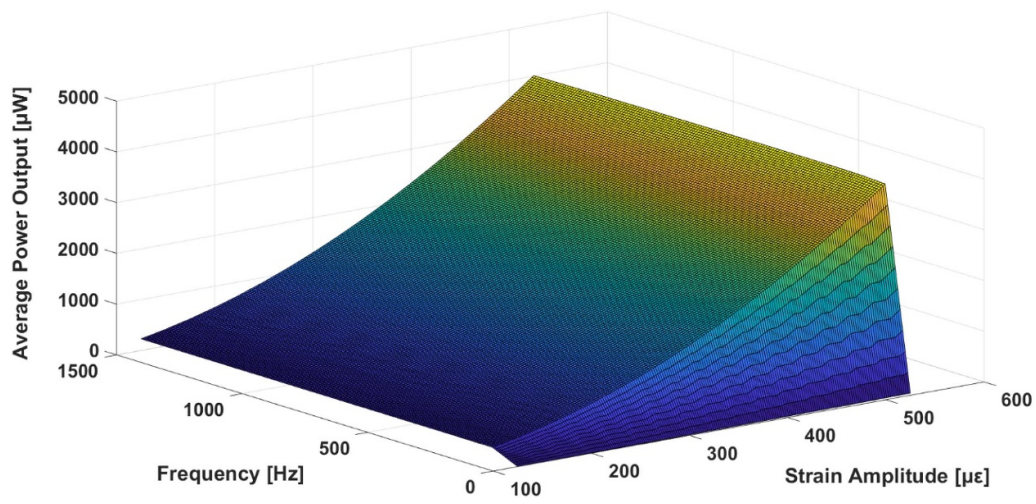


Figure 17. Average power output contour for the MPEH at high frequencies.

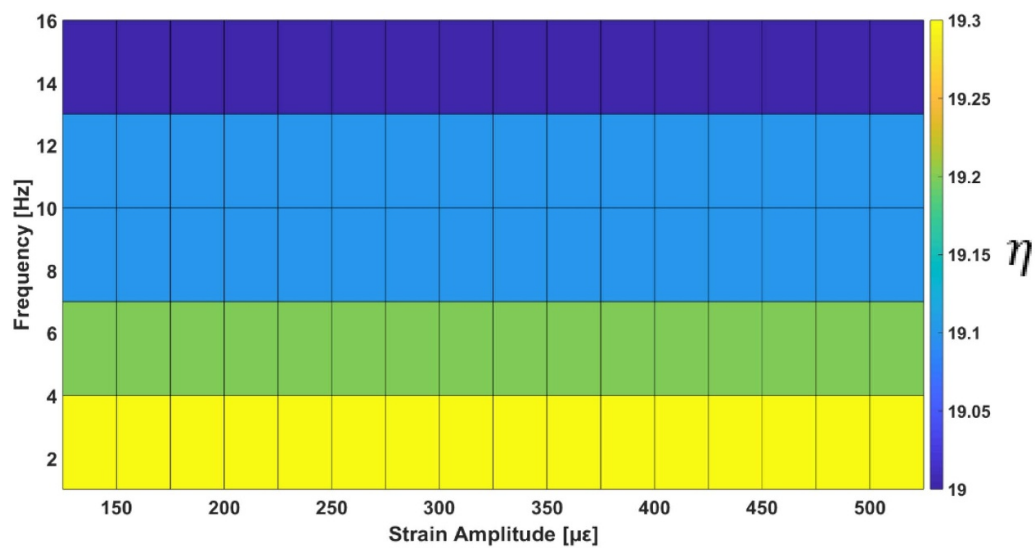


Figure 18. Distribution of the power gain factor for the MPEH.

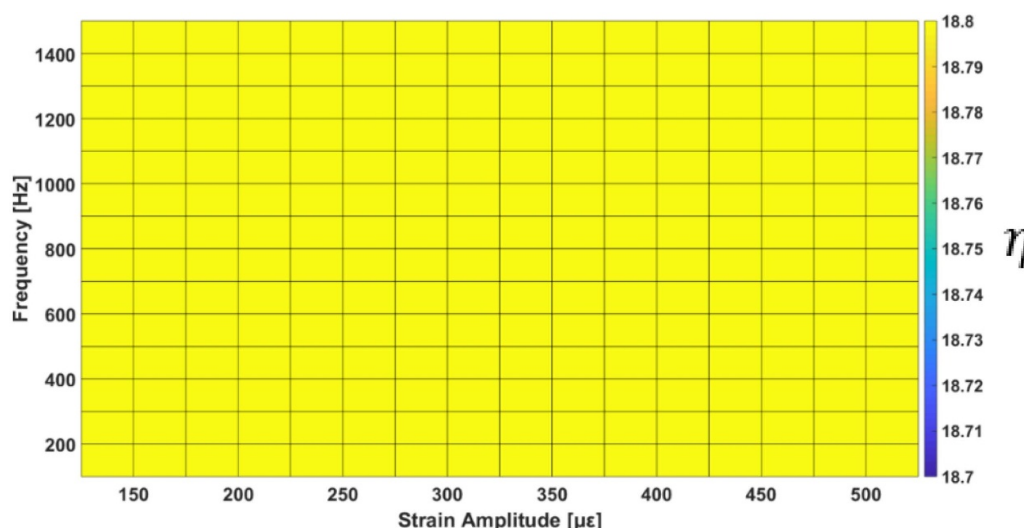


Figure 19. Distribution of the power gain factor for the MPEH.

6. Conclusion

This paper has presented a creative MetaSub for piezoelectric energy harvesting systems. The MetaSub adds an auxetic (negative Poisson's ratio) capability to a kirigami framework which enables strain energy harvester device to reach a significant planar stretchability. The FEA demonstrated that the MPEH has enhanced the PEH performance in terms of generating power with a power multiplication up to 19.3 at low level of frequencies (1–4 Hz) and sinusoidal strain amplitudes (50–550 $\mu\epsilon$). A computational parametric study has been performed on the MPEH model to optimize its material properties and dimensions. The analytical validation has been carried out to verify the numerical results. To analyze the performance of the MPEH, different combinations of both low and high frequencies excitation were investigated. The MPEH generated 18.1 time more power than an equivalent conventional PEH at high frequencies (100–1500 Hz). The simulation results indicate that the power gain factor remains approximately constant when the excitation is increased. This means the magnification factor is a function of the MetaSub geometry, and is not dependent on the excitation conditions. The power enhancement of the MPEH has been evaluated with two experimental tests for a conventional and auxetic PEHs. While the MetaSub makes a significant positive transverse and longitudinal stress in the PZT-element of device, the MetaSub still experiences a low stress distribution compared to the equivalent plain substrate of a conventional PEH. This enables the MPEH to use 82.6% of PZT capacity, which is a significant enhancement in efficiency in comparison with the conventional PEH, which uses only 7.1% of PZT capacity. At different combinations of the low level of excitation, the MPEH can potentially generate an electric power output up to 3000 μW , which easily satisfies the minimum electric power required as a sensor node for WSNs. The MPEH is easy to manufacture, and can be applied to any strain-induced structures as an appropriate sustainable power supply for portable and WSNs of autonomous SHM systems.

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